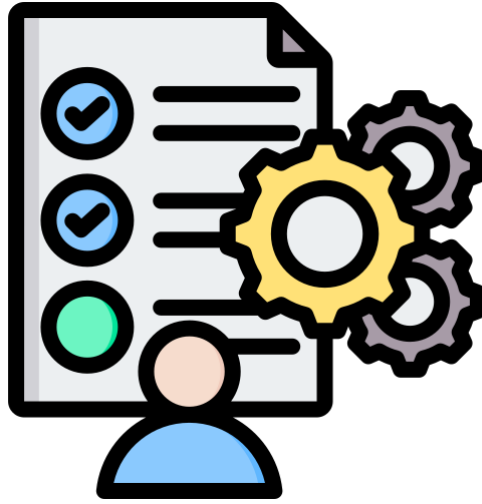




PROJECT MANAGEMENT

2025

FINAL BANK

(Solved)



BY AB & AM & MR & AE

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Choose the correct answer:

1.	Many courses in computer science and electronic engineering focus on aspects and on the notations to develop good software.				
	a) Theoretical	b) Technical	c) Efficient	d) None	
2.	To build good software, a well-defined and managed organizes the activities in an efficient and controlled way.				
	a) Theoretical	b) Process	c) A & B	d) None	
3.	A project is a endeavor undertaken to create a <i>unique</i> product, service, or result.				
	a) Temporary	b) executions	c) stakeholder	d) None	
4.	 during which the project infrastructure and the project's goals are drafted .				
	a) Initiating	b) Planning	c) Executing	d) Closing	
5.	 during which project goals are refined, activities identified and scheduled, and many other support activities are properly planned .				
	a) Initiating	b) Planning	c) Executing	d) Closing	
6.	 during which the actual work takes place. Running in parallel, a <u>monitoring phase</u> measures the progress and raises flags <u>when plans and reality disagree</u> .				
	a) Initiating	b) Planning	c) Executing	d) Closing	
7.	 where the project outputs are handed out and the project is closed .				
	a) Initiating	b) Planning	c) Executing	d) Closing	
8.	 Set of related projects managed in a <i>coordinated</i> way in order to <u>achieve some sort of benefit</u> .				
	a) Subprojects	b) Project and Program Management	c) Portfolios and Portfolio Management	d) None	
9.	 Collection of unrelated projects or programs and other work grouped to facilitate management and meet strategic objectives				
	a) Subprojects	b) Project and Program Management	c) Portfolios and Portfolio Management	d) None	
10.	Which one of the following activities spans all the stages of a software development life cycle (SDLC)?				
	a) Design	b) Testing	c) Coding	d) Project management	
11.	Types of application development				
	a) One-offs	b) Off-the-shelf	c) Customized off-the-shelf	d) All of the above	
12.	 systems specifically created for a client .				
	a) One-offs	b) Off-the-shelf	c) Customized off-the-shelf	d) All of the above	
13.	 to fill the need of a large set of users .				
	a) One-offs	b) Off-the-shelf	c) Customized off-the-shelf	d) All of the above	
14.	 standardized systems which require a <i>significant amount of</i> customization to be used in an organization like an ERP system.				
	a) One-offs	b) Off-the-shelf	c) Customized off-the-shelf	d) All of the above	



15.	_____ a temporary endeavor undertaken to create a unique product, service, or result .			
	a) Project	b) Project stakeholder	c) Portfolio Management	d) Management
16.	_____ is a <i>transient initiative</i> <u>initiated</u> to generate a distinctive product, service, or outcome			
	a) Project	b) Project stakeholder	c) Portfolio management	d) Management
17.	 obtain goals and <u>terminate</u> .			
	a) Project	b) Operational Work	c) Work	d) None
18.	_____ Achieves objectives and conclude.			
	a) Project	b) operational work	c) Work	d) Management
19.	 sustain the business.			
	a) Project	b) Operational work	c) A & B	d) None
20.	 is any <i>individual</i> or an <i>organization</i> that is actively involved in a <i>project</i> .			
	a) Project Stakeholder	b) Work	c) Project	d) Agile
21.	 whose <u>interest</u> might be affected (positively or negatively) <i>as a result of</i> project execution or completion.			
	a) Project Stakeholder	b) Work	c) Project	d) Agile
22.	 is any <i>individual</i> or an <i>organization</i> actively engaged in a project or whose <u>interest</u> may be influenced, either positively or negatively , by the execution or completion of the project.			
	a) Project	b) Portfolio Management	c) Project Stakeholder	d) Management
23.	 is the <i>integration of</i> management techniques to software development.			
	a) Software crisis	b) Software Project Management	c) Software Development Framework	d) Project Manager
24.	_____ is the fusion of management techniques with <u>software development</u>			
	a) Software project management	b) Software crisis	c) Software Development	d) Data-centric
25.	_____ describe what we <i>intend to</i> <u>achieve</u> with the project and the main characteristics of the project and its outputs.			
	a) Project goals and requirements	b) Assumptions and constraints	c) Project output and control points	d) Project goals
26.	_____ outline what we aim to accomplish with the project and the key features of the project and its deliverables.			
	a) Project goals and requirements	b) Assumptions and constraints	c) Project outputs	d) Control point
27.	_____ The process which brings to the project approval is more or less structured according to the practices of the performing organization.			
	a) Project Approval Process	b) Project goals (project scope)	c) Assumptions	d) Project goals
28.	_____ is the document that allows to formally authorize a project and to link it to the organization's goals.			
	a) Feasibility study	b) Project Approval process	c) Project goals	d) Design



29.	_____ is the document that formally authorizes a project and links it to the organization's goals.			
	a) Feasibility study	b) Project Approval process	c) project goal	d) Design
30.	The need for Software Project Management has its root in the sixties (1960) , in the days of the when practitioners recognized the <u>increasing complexity</u> of delivering software products meeting the specifications.			
	a) Software crisis	b) Software Project Management	c) Software Development Framework	d) Project Manager
31.	 are we building the <u>right</u> system?			
	a) Validation	b) Verification	c) Valid	d) Not Valid
32.	 did we build the system <u>right</u> ?			
	a) Validation	b) Verification	c) Valid	d) Not Valid
33.	 type of testing the interaction between two components .			
	a) Unit testing	b) Integration testing	c) Usability testing	d) System testing
34.	 type of testing that has a piece of code , such as a class.			
	a) Unit testing	b) Integration testing	c) Usability testing	d) System testing
35.	 type of testing the system behaves as expected and implements correctly all the requirements.			
	a) Unit testing	b) Integration testing	c) Usability testing	d) System testing
36.	checks if the user experience and interaction are intuitive, effective, and satisfying .			
	a) Unit testing	b) Integration testing	c) Usability testing	d) System testing
37.	Employed to minimize the likelihood of human errors , especially in safety-critical systems is			
	a) Unit testing	b) Integration testing	c) Usability testing	d) System testing
38.	 type of testing verifying whether the user experience and interaction is intuitive, effective, and satisfying			
	a) Unit testing	b) Integration testing	c) Usability testing	d) System testing
39.	We have approaches to <u>deploy a new system</u> .			
	a) 3	b) 4	c) 5	d) 6
40.	_____ is the duty of taking ownership of decisions made or failed to make and their consequences.			
	a) Responsibility	b) Respect	c) Fairness	d) Honesty
41.	_____ the duty of treating with respect the resources assigned to us, such as people, money, reputation, environment, and so on			
	a) Responsibility	b) Respect	c) Fairness	d) Honesty
42.	_____ the duty of taking decision impartially and objectively .			
	a) Responsibility	b) Respect	c) Fairness	d) Honesty
43.	_____ is the duty of acting in a <u>truthful</u> manner.			
	a) Responsibility	b) Respect	c) Fairness	d) Honesty
44.	Installing the new system and making it operational.			
	a) Deployment	b) Implementation	c) Operations and Maintenance	d) design



45.	 approach is an approach where the new system <u>replaces</u> the old one.			
	a) Cutover	b) Parallel	c) Piloting	d) Phased
46.	 approach is an approach where the old and the new system operate <u>simultaneously</u> for a period.			
	a) Cutover	b) Parallel	c) Piloting	d) Phased
47.	 approach is an approach where the old and the new system operate <u>concurrently</u> for a certain duration.			
	a) Cutover	b) Parallel	c) Piloting	d) Phased
48.	involves that the new system is installed for a <u>restricted number</u> of users or for a particular (specific) business unit.			
	a) Cutover	b) Parallel	c) Piloting	d) Phased
49.	 the new system is installed for a limited number of users or for a specific business unit.			
	a) Cut-over	b) Parallel Approach	c) Piloting	d) Phased Approach
50.	 functions are rolled out incrementally.			
	a) Cutover	b) Parallel	c) Piloting	d) Phased
51.	Some of the Software project management characteristics are.....			
	a) Intangible	b) Uniquely flexible	c) One-off	d) All of the above
52.	Software project management is interesting and challenging because The product is			
	a) Intangible	b) Tangible	c) May be both	d) None
53.	 is the type of maintenance, if relative to <u>fixing an issue</u> discovered after the release of the system			
	a) Corrective	b) Preventive	c) Adaptive	d) Perfective
54.	 is the type of maintenance, if relative to <u>fixing an issue</u> discovered, but not occurred (or, at least, signaled by users)			
	a) Corrective	b) Preventive	c) Adaptive	d) Perfective
55.	 is the type of maintenance, if relative to adapt a system to <u>changed external conditions</u> .			
	a) Corrective	b) Preventive	c) Adaptive	d) Perfective
56.	 is the type of maintenance, if relative to improve some characteristics of a system, like, for instance, performances.			
	a) Corrective	b) Preventive	c) Adaptive	d) Perfective
57.	 factor Is considered the <u>important factor</u> while the deployment phase that the people are ready to use the system.			
	a) Human	b) Data	c) Hardware	d) All of the above
58.	 factor is considered the important factor while the deployment phase that all the data which is needed for the system to run available to the new software.			
	a) Human	b) Data	c) Hardware	d) All of the above
59.	Is considered the important factor while the deployment phase that all interfaces ready and functional.			
	a) Human	b) Data	c) Hardware	d) All of the above
60.	 integration of systems <i>performing similar</i> .			
	a) Horizontal	b) Vertical	c) Hybrid	d) None



61.	 integration of systems <i>automating different steps of a procedure.</i>							
	a)	Horizontal	b)	Vertical	c)	Hybrid	d)	None
62.	The Managing Software Evolution is.....							
	a)	Project Development team	b)	Project Testing team	c)	Employees	d)	All of the above
63.	The..... generated by the project.							
	a)	Value	b)	Risks	c)	Agile	d)	None
64.	The associated to the project.							
	a)	Value	b)	Risks	c)	Agile	d)	None
65.	The following may be the risks of the project expect							
	a)	Resources	b)	Timing	c)	Technical uncertainties	d)	Sustainability
66.	The following may be the value of the project expect							
	a)	Value	b)	Sustainability	c)	Strategic Objectives	d)	Timing
67.	 measures the positive and negative outcomes of a project and <i>its outputs</i> .							
	a)	Project Risks	b)	Project Value	c)	Direct and Indirect Value	d)	Sustainability
68.	refers to the capacity of sustaining the project and its outputs after the project end.							
	a)	Project Risks	b)	Project Value	c)	Direct and Indirect Value	d)	Sustainability
69.	 Measures how important and relevant a project is for the performing organization.							
	a)	Alignment with the strategic objectives	b)	Sustainability	c)	Direct and Indirect Value	d)	None
70.	The techniques to assess value and risks are							
	a)	Payback period	b)	Return of Investment (ROI)	c)	Net Present Value (NPV)	d)	All of the above
71.	 Development by steps and in increments (necessary to keep a project under scope).							
	a)	Progressive elaboration	b)	Project and Program Management	c)	Application Development	d)	None
72.	 is one of the project characteristics that the development is done in steps incrementally .							
	a)	Progressive elaboration	b)	Unique products	c)	Temporary	d)	Resource-constrained
73.	Project management context includes							
	a)	Subprojects	b)	Project and Program Management	c)	Portfolio and Portfolio Management	d)	All of the above
74.	Software project management is <i>interesting and challenging</i> because Size and complexity are exponentially.							
	a)	Increasing	b)	Decreasing	c)	A & B	d)	None
75.	verification and validation (V&V) are included mostly in the phase.							
	a)	Analysis	b)	Design	c)	Testing	d)	Implementation



76.	some of the types of software development projects							
	a)	Application Development	b)	System Re-engineering	c)	System Integration	d)	All of the above
77.	 change the way in which the operational work of an organization is carried out to achieve some strategic goal (e.g., improve quality, become more efficient)							
	a)	Application Development	b)	System Re-engineering	c)	System Integration	d)	Installation and Training Services
78.	 developing an application (desktop, web, mobile, embedded)							
	a)	Application Development	b)	System Re-engineering	c)	System Integration	d)	Consulting Services
79.	 <i>automating</i> the information flow among the systems of an organization.							
	a)	Application Development	b)	System Re-engineering	c)	System Integration	d)	Consulting Services
80.	 Typically asked to gain a know-how outside a company's core competence.							
	a)	Application Development	b)	System Re-engineering	c)	System Integration	d)	Consulting Services
81.	Services related to the installation or training on specific software systems.							
	a)	Application Development	b)	System Re-engineering	c)	Installation and Training Services	d)	Consulting Services
82.	 is a revenue model in open source development.							
	a)	Application Development	b)	System Re-engineering	c)	Installation and Training Services	d)	Consulting Services
83.	We have Types of systems integration services.							
	a)	2	b)	3	c)	4	d)	5
84.	The group performing the work are considered as stakeholders.							
	a)	internal (Project team member)	b)	External	c)	In-between	d)	None
85.	Team members <u>directly</u> involved in the <u>project management</u> are considered as stakeholders.							
	a)	internal (Project management team)	b)	External	c)	In-between	d)	None
86.	Person or organization that will use the result of a project are considered as stakeholders.							
	a)	Internal	b)	External	c)	In-between (Customer/User)	d)	None
87.	Person or group providing the financial resources are considered as stakeholders.							
	a)	Internal	b)	External	c)	In-between (Sponsor)	d)	None
88.	The organization mostly involved in the project is considered as stakeholders.							
	a)	Internal	b)	External	c)	In-between (Performing Organization)	d)	None
89.	 people or groups not directly related to the project who could influence the course of a project.							
	a)	Internal	b)	External (Influencers)	c)	In-between	d)	None



90.	 the members of the team directly involved in project management.							
	a)	Sponsor	b)	Performing Organization	c)	Project Management team	d)	None
91.	 person or group providing the financial resources							
	a)	Sponsor	b)	Performing Organization	c)	Project Management team	d)	None
92.	 Person responsible for managing the project and <i>stakeholders' expectations</i>							
	a)	Quality control	b)	Project Manager	c)	Product owner	d)	Scrum master
93.	<i>Different processes</i> put different on each activity or define the in which these activities can be performed.							
	a)	Order, Emphasis	b)	Emphasis, Order	c)	Cost, changes	d)	None
94.	 is a progressive refinement that <u>moves from concept to operations</u> through some phases.							
	a)	Software project management	b)	Software crisis	c)	Software Development	d)	None
95.	 forming a <u>shared view</u> about the characteristics of the system to build							
	a)	Requirements Management	b)	Design	c)	Implementation	d)	None
96.	Free or <u>structured text</u> describing the functions and other properties of a system							
	a)	List of requirements	b)	Use case diagrams	c)	User stories	d)	None
97.	Diagrams describing the interaction between users and the system . Textual description of the interaction as a <u>sequence of steps</u>							
	a)	List of requirements	b)	Use case diagrams	c)	User stories	d)	None
98.	Structured textual descriptions of the user functions . Syntax: As a [user] I want to do [this] because of [that].							
	a)	List of requirements	b)	Use case diagrams	c)	User stories	d)	None
99.	 understanding how users will interact with the system							
	a)	User-centered analysis	b)	User-centered design	c)	A & B	d)	None
100.	 specifying how users will actually interact with the system							
	a)	User-centered analysis	b)	User-centered design	c)	A & B	d)	None
101.	 The behavior is not specified for certain cases and situations							
	a)	Inconsistencies	b)	Incompleteness	c)	Duplicates	d)	Deployment
102.	 the same requirements are described twice (<i>possibly in different ways</i>)							
	a)	Inconsistencies	b)	Incompleteness	c)	Duplicates	d)	Deployment
103.	 models the way in which an organization works.							
	a)	Business process modeling	b)	Business process re-engineering	c)	Business modeling	d)	Business development



104.	 plans the way in which an organization works, to make its operations more efficient (“as is” and “to be”)							
	a)	Business process modeling	b)	Business process re-engineering	c)	Business modeling	d)	Business development
105.	 Composition of data processing units and Focus I/O specification							
	a)	Layered/ Hierarchical	b)	Pipe and filter	c)	Data-centric	d)	Client-server
106.	Hierarchy of components and Focus control and information flow; block responsibilities							
	a)	Layered/ Hierarchical	b)	Pipe and filter	c)	Data-centric	d)	Client-server
107.	Many web applications and many desktop applications use the..... architectural style.							
	a)	Layered/ Hierarchical	b)	Pipe and filter	c)	Data-centric	d)	Client-server
108.	Data-centric is using the Architecture							
	a)	MVVM	b)	MVC	c)	VM	d)	None
109.	 Server (main functions) and clients (requesting services)							
	a)	Layered/ Hierarchical	b)	Pipe and filter	c)	Data-centric	d)	Client-server
110.	the acronym V&V is part of							
	a)	quality management	b)	quality control	c)	A & B	d)	None
111.	Use of coding and documentation standards is the phase of							
	a)	Design	b)	Implementation	c)	Analysis	d)	Validation
112.	CPM stands for.....							
	a)	Critical Path Method	b)	control path Method	c)	Critical Pass Method	d)	None
113.	PERT stands for							
	a)	Program Evaluation and Review Technique	b)	Process Evaluation and Review Technique	c)	Program Evaluation and Retrieve Technique	d)	None
114.	SWOT stands for.....							
	a)	Strengths, Weaknesses, Opportunities, Threats	b)	Strengths, Weaknesses, Opportunities, Trust	c)	Strengths, Weaknesses, Options, Threats	d)	Strengths, Weaknesses, Opportunities, Treatment
115.	A _____ is also known as three-point estimation.							
	a)	PERT	b)	Function points	c)	COCOMO	d)	None
116.	Ensuring the system runs smoothly .							
	a)	Analysis	b)	implementation	c)	operations and maintenance	d)	design



117.	Ensuring the smooth operation of the system is.....							
	a)	Analysis	b)	implementation	c)	operations and maintenance	d)	design
118.	_____ is the time taken to gain a financial return equal to the original investments.							
	a)	Payback Period	b)	Return on Investment (ROI)	c)	Net Present Value (NPV)	d)	Internal Rate of Return
119.	_____ calculates the <u>average annual profit</u> and transforms it into a percentage of the total investments .							
	a)	Payback Period	b)	Return on Investment (ROI)	c)	Net Present Value (NPV)	d)	Internal Rate of Return
120.	_____ is a method used to evaluate the profitability of an investment by calculating the difference between the present value of cash inflows and the present value of cash outflows over time.							
	a)	Payback Period	b)	Return on Investment (ROI)	c)	Net Present Value (NPV)	d)	Internal Rate of Return
121.	_____ is the discount rate at which NPV = 0 , it is used in investment planning to evaluate the profitability of a project .							
	a)	Payback Period	b)	Return on Investment (ROI)	c)	Net Present Value (NPV)	d)	Internal Rate of Return (IRR)
122.	If IRR _____ required rate of return, the project is considered profitable .							
	a)	More than	b)	Less than	c)	Equal to	d)	None
123.	If IRR _____ required rate of return, the project is considered unprofitable .							
	a)	More than	b)	Less than	c)	Equal to	d)	None
124.	A chart representing project tasks against a calendar in a horizontal bar format is known as _____.							
	a)	Milestone	b)	Goal	c)	Gantt chart	d)	PERT chart
125.	_____ is a horizontal bar chart that shows project tasks against a calendar .							
	a)	Milestone	b)	Goal	c)	Gantt chart	d)	PERT chart
126.	A _____ is a hierarchical decomposition of the work to be executed by the project team to accomplish projects objectives and create the required deliverables.							
	a)	WBS	b)	PERT	c)	Effort	d)	None
127.	WBS Types are _____							
	a)	Product WBS	b)	Process WBS	c)	Hybrid WBS	d)	All of the above
128.	_____ It develops according to the structure of the outputs that need to be produced. It can start from a Product Breakdown Structure , when defined.							
	a)	Product WBS	b)	Process WBS	c)	Hybrid WBS	d)	None
129.	_____ It develops according to the phases in which a project is organized . For instance: Requirements, Analysis, Design, Testing.							
	a)	Product WBS	b)	Process WBS	c)	Hybrid WBS	d)	None
130.	_____ It mixes process and product where life-cycle phases at higher levels ; component at lower levels .							
	a)	Product WBS	b)	Process WBS	c)	Hybrid WBS	d)	None
131.	_____ Higher levels are geographically distributed teams (e.g. NY team, Trento Team). - Lower levels collect the work under the responsibility							
	a)	Geographical WBS	b)	Product WBS	c)	Organizational WBS	d)	Process WBS



132.	_____ Higher levels are organizational units. - Lower levels collect the work which is under the responsibility of a Unit							
	a)	Geographical WBS	b)	Product WBS	c)	Organizational WBS	d)	Process WBS
133.	Each level of a WBS includes all the items that are necessary to develop the parent node. This is known as _____							
	a)	balance rule	b)	100% rule	c)	Mutually Exclusive rule	d)	dependency rule
134.	_____ make sure all work items are there							
	a)	balance rule	b)	100% rule	c)	Mutually Exclusive rule	d)	dependency rule
135.	_____ are better suited for the 100% rule .							
	a)	Product WBS	b)	Process WBS	c)	Hybrid WBS	d)	None
136.	_____ make sure there are no overlaps in the definition of the elements							
	a)	Balance rule	b)	100% rule	c)	Mutually Exclusive rule	d)	Dependency rule
137.	in a WBS, there are no overlaps between nodes. This is known as _____							
	a)	Balance rule	b)	100% rule	c)	Mutually Exclusive rule	d)	Dependency rule
138.	A _____ is an <u>element of the WBS</u> at which <u>estimation for time and costs</u> can be reliably provided .							
	a)	Work package	b)	Dependency	c)	Monitoring	d)	None
139.	A _____ helps further specify the entries of a WBS. - It might contain title, number, detailed description of the element, quantities, associated work, contractual items.							
	a)	WBS dictionary	b)	WBS database	c)	WBS file	d)	WBS lexicon
140.	A _____ is an auxiliary document containing details about work package							
	a)	WBS lexicon	b)	WBS dictionary	c)	WBS database	d)	WBS file
141.	_____ an <u>element of project work</u> that requires action to produce deliverable.							
	a)	work package	b)	Dependency	c)	Activity	d)	None
142.	What WBS construction approach has the risk of violating the mutual exclusion rule ?							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None
143.	What WBS construction approach has the Best suited for projects that are new for a company .							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None
144.	_____ the leaves of the WBS are identified and then are grouped in homogeneous items.							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None
145.	_____ the construction of the WBS proceeds from the top level down to the leaves .							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None
146.	What WBS construction approach has the risk is overlooking activities (e.g., achieving a “90% decomposition”).							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None



147.	What WBS construction approach has the Best suited for projects that are well known or whose structure is clear.							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None
148.	_____ <u>start from an existing</u> WBS, which is adapted and customized for the project at hand							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None
149.	What WBS construction approach has the risk effective when an organization standardizes the structure of its projects.							
	a)	Top-down	b)	Bottom-up	c)	By analogy	d)	None
150.	_____ how long will the activity last for.							
	a)	Duration	b)	Effort	c)	Resources	d)	none
151.	_____ is the length of time that the work is spread across.							
	a)	Duration	b)	Effort	c)	Resources	d)	none
152.	_____ how much work will the activity need to be completed .							
	a)	Duration	b)	Effort	c)	Resources	d)	none
153.	_____ amount of work required to complete an activity. A very good starting point.							
	a)	Duration	b)	Effort	c)	Resources	d)	none
154.	_____ is man-hours of work required to complete an activity.							
	a)	Duration	b)	Effort	c)	Resources	d)	none
155.	_____ <u>type and quantity of resources</u> available the activity.							
	a)	Duration	b)	Effort	c)	Resources	d)	none
156.	_____ are <u>needed</u> to carry the work out. Typically a constraints (limited).							
	a)	Duration	b)	Effort	c)	Resources	d)	none
157.	_____ can reduce duration , assuming the <u>work can be divided</u> without loss of efficiency.							
	a)	Increasing manpower	b)	Increasing duration	c)	Diminishing Returns	d)	None
158.	_____ can allow for fewer people to be assigned, but the total effort required remains the same							
	a)	Increasing manpower	b)	Increasing duration	c)	Diminishing Returns	d)	None
159.	_____ Simply adding more people (manpower) doesn't always reduce duration proportionally.							
	a)	Increasing manpower	b)	Increasing duration	c)	Diminishing Returns	d)	None
160.	_____ each activity includes some contingency time.							
	a)	Implicit padding	b)	Explicit padding	c)	React and re-plan	d)	None
161.	_____ the contingency time is explicitly modeled <u>as an activity</u> .							
	a)	Implicit padding	b)	Explicit padding	c)	React and re-plan	d)	None
162.	_____ when a delays occurs, you re-plan and redefine a new realistic schedule.							
	a)	Implicit padding	b)	Explicit padding	c)	React and re-plan	d)	None



163.	What technique that estimates duration “ <i>quick and dirty</i> ” and based on experience ? It can be applied either top-down or bottom-up.			
	a) Expert Judgement	b) PERT	c) Algorithmic	d) None
164.	What technique that estimates duration that takes into account the probabilistic nature of estimations?			
	a) Expert Judgement	b) PERT	c) Algorithmic	d) None
165.	What technique that estimates duration by measuring specific qualities of a system and applying algorithms (Function Points, COCOMO)			
	a) Expert Judgement	b) PERT	c) Algorithmic	d) None
166.	COCOMO stands of _____			
	a) Constructive Cost Modeling	b) Constructive Modeling Cost	c) Cost Modeling Constructive	d) None
167.	What technique that estimates duration based defining a function of a set of measurable characteristics ?			
	a) Expert Judgement	b) PERT	c) Algorithmic	d) None
168.	Estimation in PERT is based on the idea that estimates are _____.			
	a) uncertain	b) certain	c) Probabilistic	d) None
169.	what PERT Formula is shortest possible time to complete a task?			
	a) Optimistic (O)	b) Most likely (M)	c) Pessimistic (P)	d) None
170.	what PERT Formula is most realistic time estimate, assuming normal conditions?			
	a) Optimistic (O)	b) Most likely (M)	c) Pessimistic (P)	d) None
171.	what PERT Formula is longest possible time to complete a task?			
	a) Optimistic (O)	b) Most likely (M)	c) Pessimistic (P)	d) None
172.	_____ is the square root of variance .			
	a) Standard deviation	b) Variance	c) Task duration	d) Normal distribution
173.	_____ measure(s) how spread a population is from the average.			
	a) Variance	b) Standard deviation	c) Both A & B	d) None
174.	_____ is a bell-shaped probability distribution function			
	a) Normal distribution	b) Abnormal distribution	c) Standard deviation	d) Variance
175.	_____ estimations focus on measuring a software system in terms of its functions or functional components.			
	a) Function-Based	b) Size-Based	c) file-based	d) None
176.	_____ estimations break down the system into logical units of work that are easier to estimate, regardless of the specific technology or platform being used.			
	a) Function-Based	b) Size-Based	c) file-based	d) None
177.	what techniques Algorithmic estimations software system's physical size and then estimate the effort required based on that sized.			
	a) Function-Based	b) Size-Based	c) file-based	d) None



178.	A _____ approach estimates effort and duration based on the number of lines of code .			
	a) PERT	b) function points	c) COCOMO	d) None
179.	what technique estimates effort based on functional characteristics of the system?			
	a) PERT	b) Function Point	c) COCOMO	d) None
180.	Function point estimation is an example of _____ estimations.			
	a) Function-based	b) Size-based	c) Both	d) None
181.	COCOMO is an example of _____ estimations.			
	a) Function-based	b) Size-based	c) Both	d) None
182.	_____ states that work expands to fill the time available for its completion.			
	a) Parkinson's law	b) Student's syndrome	c) overconfidence	d) None
183.	_____ refers to planned procrastination to the last possible time before the deadline.			
	a) Parkinson's law	b) Student's syndrome	c) overconfidence	d) None
184.	A _____ constraint Can be broken without violating the logic of the project.			
	a) Hard dependencies	b) Soft dependencies	c) Identify dependencies	d) none
185.	A _____ constraint Cannot be broken without violating the logic of the project.			
	a) Hard dependencies	b) Soft dependencies	c) Identify dependencies	d) none
186.	Dependencies that can be broken without violating the logic of the project are called _____ dependencies			
	a) hard	b) soft	c) reflexive	d) transitive
187.	Breaking _____ dependencies can cause violation of the project logic.			
	a) hard	b) soft	c) reflexive	d) transitive
188.	The Task dependencies Relationships, B cannot start till A finishes is _____			
	a) Finish-to-Start (FS)	b) Start-to-Start (SS)	c) Finish-to-Finish (FF)	d) Start-to-Finish (SF)
189.	The Task dependencies Relationships, B cannot start till A starts is _____			
	a) Finish-to-Start (FS)	b) Start-to-Start (SS)	c) Finish-to-Finish (FF)	d) Start-to-Finish (SF)
190.	The Task dependencies Relationships, B cannot finish till A finishes is _____			
	a) Finish-to-Start (FS)	b) Start-to-Start (SS)	c) Finish-to-Finish (FF)	d) Start-to-Finish (SF)
191.	The Task dependencies Relationships, B cannot finish till A starts (rare) is _____			
	a) Finish-to-Start (FS)	b) Start-to-Start (SS)	c) Finish-to-Finish (FF)	d) Start-to-Finish (SF)
192.	_____ delay introduced by the dependency is positive A Positive some time passes between the two tasks.			
	a) Lag	b) Lead	c) Slack	d) None
193.	_____ the duration of the dependency is negative (the activities partially overlap).			
	a) Lag	b) Lead	c) Slack	d) None
194.	_____ are made to clearly mark "phase" transitions			
	a) Milestones	b) Tasks	c) Activities	d) Dependencies



195.	The _____ is the longest sequence of dependent tasks in a project that determines the shortest possible duration to complete the project.							
	a)	Gantt chart	b)	WBS	c)	Critical path	d)	Critical Task
196.	The path in which all activities have zero slack is called_____							
	a)	Gantt chart	b)	WBS	c)	Critical path	d)	Critical Task
197.	_____ is the amount of time an activity can be delayed without affecting the end of a project.							
	a)	Lag	b)	Lead	c)	Slack	d)	None
198.	In Network Diagram, _____ diagrams represent activities on the nodes							
	a)	Precedence (AON)	b)	Arrow (AOA)	c)	Arc	d)	None
199.	In Network Diagram, _____ diagrams represent activities on the edges							
	a)	Precedence (AON)	b)	Arrow (AOA)	c)	Arc	d)	None
200.	AON Standard of _____							
	a)	Activity On Node	b)	Active On Node	c)	All Optical Network	d)	None
201.	AOA Standard of _____							
	a)	Activity On Node	b)	Activity On Arrow	c)	All Optical Network	d)	None
202.	_____ is the earliest date at which we can start an activity without breaking any dependency on the plan.							
	a)	Earliest start	b)	Earliest end	c)	Latest start	d)	Latest end
203.	_____ is the earliest date at which we can finish an activity without breaking any dependency on the plan.							
	a)	Earliest start	b)	Earliest end	c)	Latest start	d)	Latest end
204.	_____ is the latest date at which we can start an activity without delaying or moving any other activity.							
	a)	Earliest start	b)	Earliest end	c)	Latest start	d)	Latest end
205.	_____ is the latest date at which we can finish an activity without delaying or moving any other activity							
	a)	Earliest start	b)	Earliest end	c)	Latest start	d)	Latest end
206.	A _____ determines the earliest start and earliest end dates of each activity in the plan.							
	a)	forward pass	b)	backward Pass	c)	Critical Task	d)	None
207.	A _____ determines the latest start and latest end dates of each activity in the plan.							
	a)	forward pass	b)	backward Pass	c)	Critical Task	d)	None
208.	_____ is the process of introducing soft constraints in a plan to ensure that no resource is over- allocated.							
	a)	Resource leveling	b)	Critical Path Method	c)	Resource Allocation	d)	Resource Smoothing
209.	The task has to be started at the latest possible date, given any other constraint is satisfied called_____							
	a)	As late as possible	b)	Start no later than	c)	Finish no later than	d)	None
210.	A _____ task has to start on a specific date.							
	a)	Must Start on	b)	Must finish on	c)	Must Start off	d)	Must finish off
211.	A _____ the task has to end on a specific date.							
	a)	Must Start on	b)	Must finish on	c)	Must Start off	d)	Must finish off



212.	In Resource leveling _____ the task cannot start earlier than a specific date, but it is perfectly fine if it starts later than the set date.				
	a)	Start no earlier than	b)	Start no later than	c) End no earlier than d) Finish no later than
213.	In Resource leveling _____ the task has to start no later than a given date, but it is perfectly fine if it starts earlier than the set date.				
	a)	Start no earlier than	b)	Start no later than	c) End no earlier than d) Finish no later than
214.	In Resource leveling _____ the task cannot finish earlier than a specific date, but it is perfectly fine if it ENDS later than the set date				
	a)	Start no earlier than	b)	Start no later than	c) End no earlier than d) Finish no later than
215.	In Resource leveling _____ the task cannot finish later than a given date, but it is perfectly fine if it ends before the set date.				
	a)	Start no earlier than	b)	Start no later than	c) End no earlier than d) Finish no later than
216.	_____ is a method for shortening the project duration by reducing the time of one or more of the critical project activities to less than its normal activity time.				
	a)	Project crashing	b)	Project Management	c) Fast tracking d) None
217.	Reducing the project duration can be achieved by reducing the time of one or more of the <u>critical path</u> activities, this Known as _____				
	a)	CCM	b)	Fast tracking	c) Project crashing d) None
218.	Reducing the project duration by reducing the time of one or more of the critical path activities is known as _____.				
	a)	CCM	b)	fast tracking	c) project crashing d) None
219.	_____ is based on the fact that deliverables of activities are <u>incrementally produced (and refined)</u> during the execution of the activity.				
	a)	Project crashing	b)	Fast tracking	c) Management d) None
220.	_____ dependencies derive from the project logic or from external dependencies .				
	a)	Hard	b)	Soft	c) Identify d) None
221.	_____ dependencies derive from the discretionary choices or from resource availability and learning .				
	a)	Hard	b)	Soft	c) Identify d) None
222.	Fast tracking is performed by _____				
	a)	Overlapping Activities	b)	Allocating more resources to the activities	c) Reducing the work scope d) None
223.	the main risk with fast tracking is _____				
	a)	increasing the duration of the project	b)	Increasing the cost of the project	c) Rework might be necessary d) None
224.	if A And B are connected by an FS dependency with a <u>lag time of 3 days</u> , it means that _____				
	a)	B can start three days after A has finished	b)	B could start 3 days before A ends	c) B starts directly after A ends d) None



225.	The following factors can affect the start and finish date of a project EXCEPT _____						
	a)	Holidays	b)	Task dependencies	c)	constraints	d) Quality standard
226.	Triple Constraint Triangle refers to _____, _____ and _____.						
	a)	Quantity, Budget, Time	b)	Quality, Time, Cost	c)	Quantity, Cost, Time	d) None
227.	_____ helps understand whether the project will be in the manager's comfort zone or not						
	a)	Client	b)	Performing organization	c)	Project manager	d) Programmer

True/False:

1.	Responsibility is the duty of treating with <u>respect</u> the resources assigned to us, such as people, money, reputation, environment, and so on. (Respect)	F
2.	Respect is the duty of taking ownership of decisions made or failed to make and their consequences. (Responsibility)	F
3.	<u>Fairness</u> is the duty of taking decisions <u>impartially</u> and <u>objectively</u> .	T
4.	<u>Honesty</u> is the duty of acting in a <u>truthful</u> manner.	T
5.	<u>Mars Climate Orbiter bug</u> shows the importance of unit testing (integration)	F
6.	An advantage of <u>list of requirements</u> is that it's simple to draft and distribute.	T
7.	A disadvantage of <u>list of requirements</u> is that the format can be used to keep track of changes. (advantage)	F
8.	<u>Unit</u> testing scope: a piece of <u>code</u> , such as class.	T
9.	<u>Integration</u> testing scope: the <u>interaction</u> between <u>two components</u> .	T
10.	<u>System</u> testing scope: the <u>system</u> behaves as expected and implements correctly all the requirements.	T
11.	<u>Usability</u> testing scope: verifying whether the <u>user experience</u> and interaction is intuitive, effective and satisfying, and is used to reduce the probability of <u>human errors</u> .	T
12.	The goal of <u>operations and maintenance</u> is <u>ensuring</u> the system runs <u>smoothly</u> .	T
13.	The goal of <u>implementation</u> is <u>writing</u> the code.	T
14.	The goal of <u>system design</u> is to <u>define</u> the structure of the software to build (system architecture)	T
15.	The goal of <u>requirements structuring</u> is <u>improving maintenance</u> of <u>requirements</u> over time.	T
16.	In the <u>use case diagrams'</u> <u>disadvantages</u> , Managing <u>diagrams</u> requires a bit more work than working with text only.	T
17.	WBS is a static document does not change along life cycle of project	F
18.	WBS can be used for monitoring project	T
19.	WBS can be balanced	T
20.	WBS must be balanced (doesn't have to)	F
21.	There are multiple variations of PERT.	T
22.	The critical path is path with the shortest duration.	T
23.	Any delay in any activity in the critical path will cause a delay to the data of the project.	T
24.	Fast tracking is suitable for activities that produce stable intermediate results.	T
25.	CCM uses best guesses to <u>estimate each activity in the chain</u> and adds cautions estimations at the end of each activity . (the chains)	F
26.	Work package can correspond the root of the WBS. (leaves)	F



27.	The graphical format of a WBS is similar to the table of contents of book. (textual format)	F
28.	A WBS tree does not need to be balance.	T
29.	Both effort and duration have units of time.	T
30.	To use PERT, three values have to be assigned to each activity.	T

FINAL QUESTIONS

(Duration-Effort-Manpower, PERT/CPM)

1.	Suppose that <u>1 week = 40 hours</u>, if the effort required to complete a task equals 120 man-hours and only one worker is available 50% of the time, how many weeks are required to complete the task? (1 week = 40 hours, Effort = 120 man-hours, Resources = 1 * 50%) $\text{Duration} = \frac{120}{50\%} = 240 \text{ hours} / 40 = 6 \text{ weeks}$				
	a) 3	b) 4	c) 5	d) 6	
2.	if you have 3 persons working on a task with the following percentage of availability (30%, 100%, and 70%) then the <u>man power</u> equal _____ (1 * 30% + 1 * 100% + 1 * 70% = 2)				
	a) 3	b) 2	c) 200	d) 300	
3.	if the most likely, optimistic, and pessimistic time estimates equal 30,22, and 32 days, repetitively, The PERT estimation equals _____ days. (Optimistic (O) = 22, Most Likely (M) = 30, Pessimistic (P) = 32) $T_E = \frac{O+4M+P}{6} = \frac{22 + 4(30) + 32}{6} = 29$				
	a) 26	b) 29	c) 30	d) 31	



Table 1: Dependency Table

Activity No.	Duration (Weeks)	Predecessors
1	5	
2	2	
3	4	
4	4	1
5	5	1,2
6	2	1,2
7	2	5,6
8	1	4,6

4.	Refer to table 1, what is the slack for activity 7?							
	a)	0 Weeks	b)	3 Weeks	c)	4 Weeks	d)	8 Weeks
5.	Refer to table 1, what is the slack for activity 2?							
	a)	0 Weeks	b)	3 Weeks	c)	4 Weeks	d)	8 Weeks
6.	Refer to table 1, what is LFT for activity 5?							
	a)	Week 8	b)	Week 9	c)	Week 10	d)	Week 11
7.	Refer to table 1, What is ESL for activity 5?							
	a)	Week 2	b)	Week 5	c)	Week 6	d)	Week 7
8.	Refer to Table 1, what is the total project Duration?							
	a)	5 Weeks	b)	10 Weeks	c)	12 Weeks	d)	14 Weeks

WRITTEN EXPLANATION

1. Duration, Effort, Resources:

D => المدة اللازمة لإكمال المهمة

E => الجهد المطلوب

M => عدد الأشخاص



$$D = \frac{E}{M}$$

Examples:

1 week = 40 hours

1. Effort: 40 man-hours ; Resources : 1 @ 100%

D = 40 man-hours / 1 man = 40 hours = 1 week

2. Effort: 80 hours ; Resources: 2 @ 100%

D = 80 man-hours / 2 man = 40 hours = 1 week

3. Effort : 80 hours ; Resources: 1 @ 50%

D = 80 / 50% = 160 hours = 4 weeks

هنا عدد الأشخاص واحد بس بيشغل بنسبة 50 % فهنقسم على 0.5

2. Function Point:

Function Point

Example: If you have a system with:

- 5 user inputs (low complexity), each worth 4 FP
→ $5 \times 4 = 20 \text{ FP}$
- 3 reports (high complexity), each worth 10 FP →
 $3 \times 10 = 30 \text{ FP}$
- 2 internal files (average complexity), each worth 7 FP →
 $2 \times 7 = 14 \text{ FP}$

Total function points = $20 + 30 + 14 = 64 \text{ FP}$.



3. Payback Period:

payback period العائد السنوي = Initial Investment / الاستثمار الاول / Annual Cash Inflows



- Imagine a company invests \$50,000 in new machinery that generates annual cash inflows of \$12,500.
- The payback period is calculated as follows:
$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflows}} = \frac{50,000}{12,500} = 4 \text{ years}$$
- This means the company will recoup its initial \$50,000 investment in 4 years.

4. Return of Investment (ROI):



- Profit** = Returns - Investments الاستثمار
- Annual Profit** = Profit / Duration المدة
- ROI** = Annual Profit / Investments

	Project A	Project B	Project C
Year 0	€ (50,000.00)	€ (20,000.00)	€ (15,000.00)
Year 1	€ 30,000.00	€ (10,000.00)	€ 15,000.00
Year 2	€ 30,000.00	€ 10,000.00	€ 1,000.00
Year 3	€ 1,000.00	€ 60,000.00	
Year 4	€ 1,000.00	€ 50,000.00	

العباسي

اللي بالاحمر هو الاستثمار
كل عمود عبارة عن مشروع

1. Project A (العمود الأول)

- Profit = 62000 - 50000 = 12000
- Annual Profit = 12000 / 4 = 3000
- ROI = 3000 / 50000 = 0.06 = 6%

2. Project B (العمود الثاني)

- Profit = 120000 - 30000 = 90000
- Annual Profit = 90000 / 4 = 22500
- ROI = 22500 / 30000 = 75%

3. Project C (العمود الثالث)

- Profit = 16000 - 15000 = 1000
- Annual Profit = 1000 / 2 = 500
- ROI = 500 / 15000 = 3%



5. Net Present Value (NPV):

$$NPV = \sum \frac{C_t}{(1+r)^t} - C_0$$

Where C_t = Cash inflow at time t ,

r = Discount rate, and

C_0 = Initial investment (cash outflow)

- If $NPV > 0$, the investment is considered profitable. If $NPV < 0$, the investment is not profitable.

C_t : التدفق المالي في الزمن t

r : Discount Rate سعر الخصم

C_0 : الاستثمار الأولي (التكلفة)

$NPV > 0 \rightarrow$ considered profitable

$NPV < 0 \rightarrow$ NOT profitable

Example:

Hypothesis
Discount Rate: 10%
(this is " r ")

$$\text{Discount Factor} = \frac{1}{(1 + i)^n}$$

Year (n)	Cash Flow	Discount Factor	Present Value
0	€ (35,000.00)	1.00	€ (35,000.00)
1	€ 10,000.00	0.91	€ 9,090.91
2	€ 15,000.00	0.83	€ 12,396.69
3	€ 20,000.00	0.75	€ 15,026.30
Expenditure	€ (35,000.00)		€ (35,000.00)
Gains	€ 45,000.00		€ 36,513.90
Profit	€ 10,000.00		€ 1,513.90

سالب عشان دا استثمار (مدفوع)

$$\begin{aligned}
 & -35000,00 = 35000,00 * 1 = \frac{1}{(1+0,1)^0} \quad \text{السنة 0} \\
 & + 9090,91 = 10000,00 * 0,91 = \frac{1}{(1+0,1)^1} \quad \text{السنة 1} \\
 & + 12396,69 = 15000,00 * 0,83 = \frac{1}{(1+0,1)^2} \quad \text{السنة 2} \\
 & + 15026,30 = 20000,00 * 0,75 = \frac{1}{(1+0,1)^3} \quad \text{السنة 3}
 \end{aligned}$$

$NPV = 1513.9 > 0 \Rightarrow$ considered profitable



6. PERT/CPM:

ES	Duration	EE
Task Name		
LS	slack	LE

• Duration معطي في السؤال

- خلي بالك من كل Activity مين هو dependency بتاعه
- المسألة مكونه من forward pass و backward pass

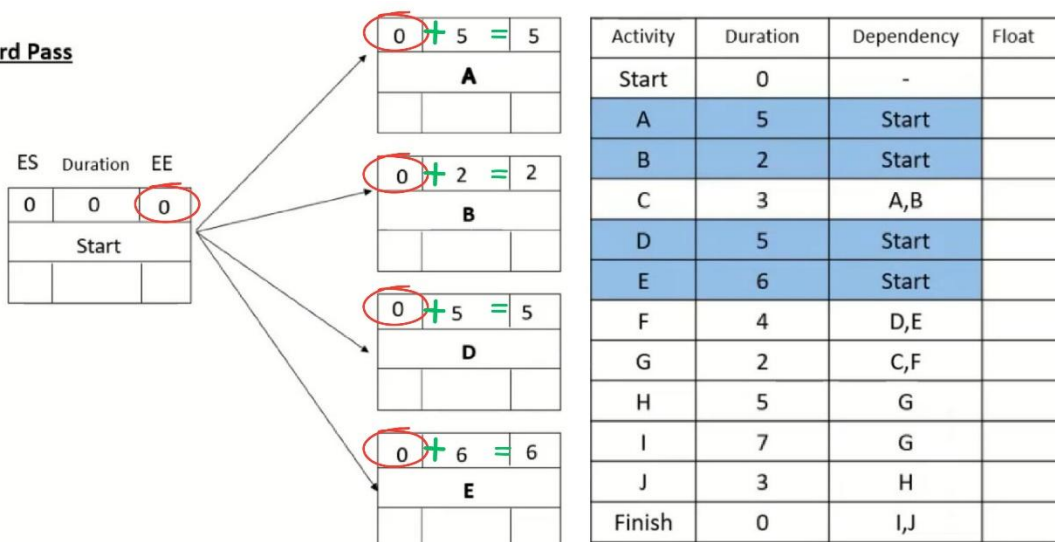
Exercise on Critical Path:

For the following table of information,

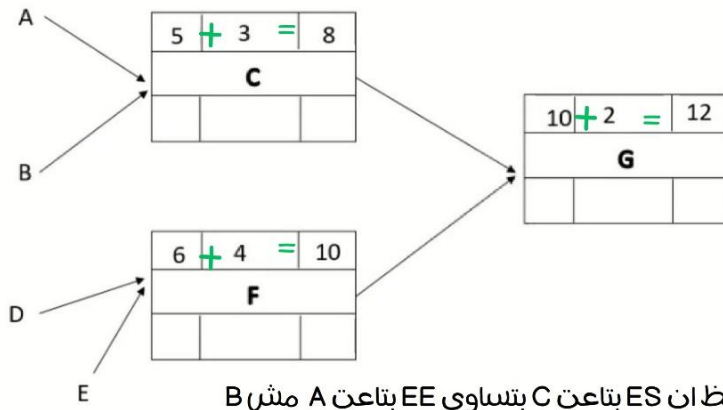
1. Draw the network diagram
2. Determine the slack for each activity
3. List the network paths
4. Determine the critical path(s)

Activity	Duration	Dependency	Slack
Start	0	-	
A	5	Start	
B	2	Start	
C	3	A,B	
D	5	Start	
E	6	Start	
F	4	D,E	
G	2	C,F	
H	5	G	
I	7	G	
J	3	H	
Finish	0	I,J	

1. forward Pass

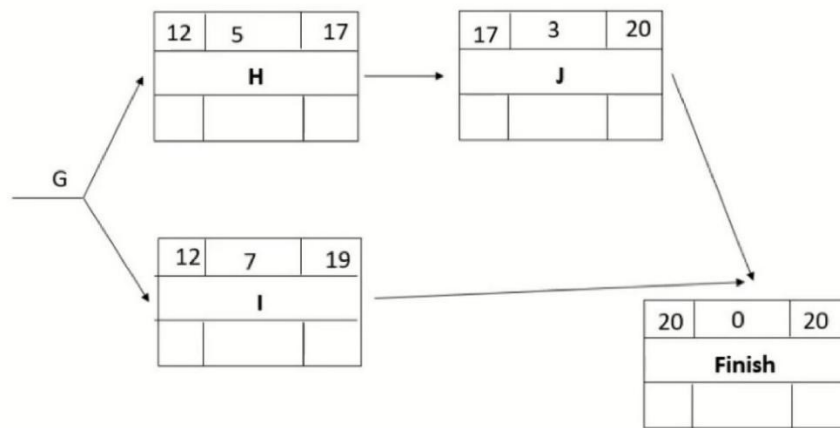


- لاحظ ان start كل حاجه فيها ب صفر
- لاحظ ان A, B, D, E كلهم ال Dependency بتاعهم هو start "اللي بالازرق"
- ال EE بتاع dependency هو ES بتاع Activity اللي معتمد عليه "اللي بالاحمر"
- ولو فيه Activity له اتنين dependency بنأخذ الاكبر
- لاحظ ان $EE = Duration + ES$ "بالاخضر"



• لاحظ ان ES بتاعت C بتساوي EE بتاعت A مش B
عشان A اكبر وكذلك في G هتلاقي F اكبر

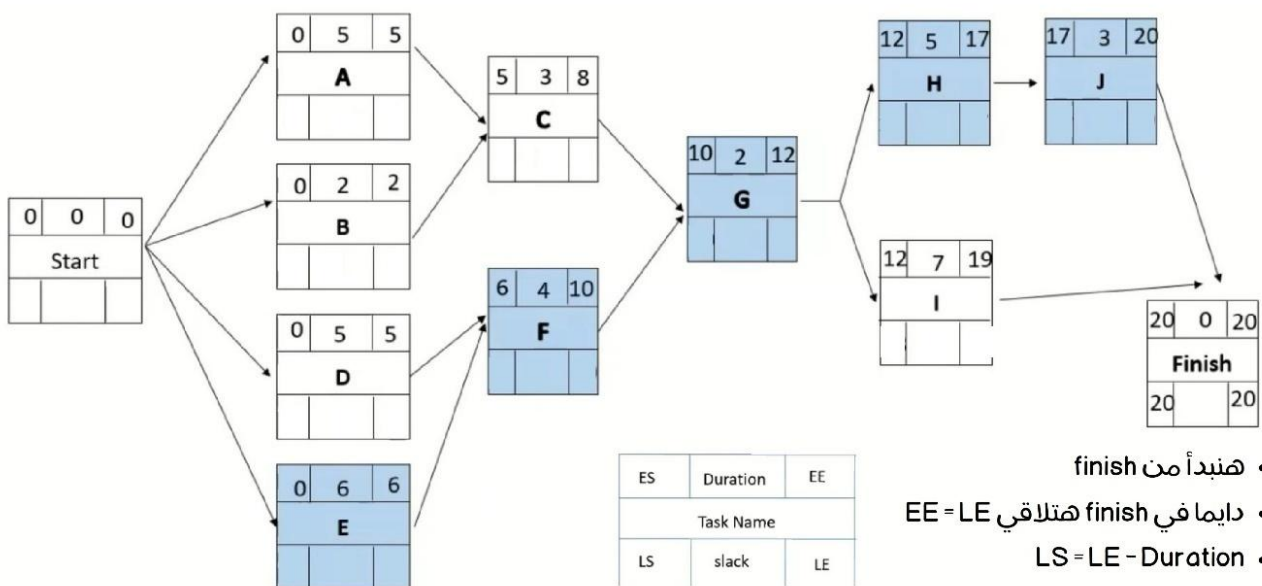
Activity	Duration	Dependency	Slack
Start	0	-	
A	5	Start	
B	2	Start	
C	3	A,B	
D	5	Start	
E	6	Start	
F	4	D,E	
G	2	C,F	
H	5	G	
I	7	G	
J	3	H	
Finish	0	I,J	



• كذا خلصنا forward pass

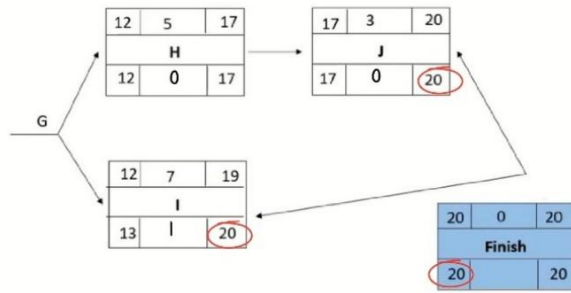
Activity	Duration	Dependency	Float
Start	0	-	
A	5	Start	
B	2	Start	
C	3	A,B	
D	5	Start	
E	6	Start	
F	4	D,E	
G	2	C,F	
H	5	G	
I	7	G	
J	3	H	
Finish	0	I,J	

backward pass



• هنبداً من finish
• دايماً في finish هتلاقي $EE = LE$
• $LS = LE - \text{Duration}$

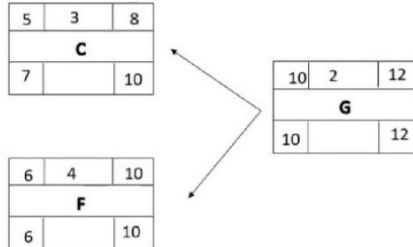




- لاحظ ان LS بتاعت Activity (في الحالة دي هي finish) بتساوي LE بتاعت dependency (اللي هي I, J) - بالاحمر

• $LS = LE - \text{Duration}$

• ملحوظة: القانون دا في Task الواحدة مش بين كذا Task (يعني LS بتاعت I بتساوي LE بتاعت I - ال Duration بتاعت I)

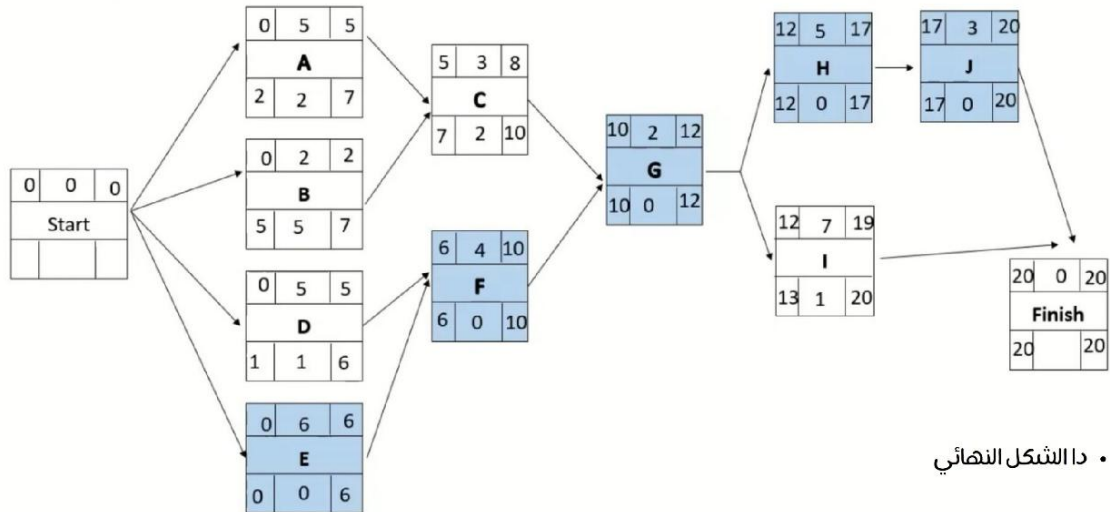


• $\text{slack} = LE - EE$ او $LS - ES$

• ملحوظة: القانون دا في Task الواحدة مش بين كذا Task

- خلي بالك وانت رايح من H و I الى G هتلاقى عندك اثنين LS $\leq$ خذ الاقل دايما يكون هو LE بتاعت G

Network Diagram & The slack for each activity



• دا الشكل النهائي

أسهل طريقة تجيب بيها ال **critical path** هي انك تشوف في الرسم التاسكات اللي عندها **slack** بـ 0 زي الرسم اللي فوق

• عشان تحسب Critical path اكبر مسار من حيث Duration

Network paths:

1. A - C - G - H - J = 18
2. A - C - G - I = 17
3. B - C - G - H - J = 15
4. B - C - G - I = 14
5. D - F - G - H - J = 19
6. D - F - G - I = 18
7. E - F - G - H - J = 20 <=====
8. E - F - G - I = 19

Critical path:

E - F - G - H - J



الاحتياط واجب

1.	The first step in creating a project is _____.			
	a) Defining the project	b) Managing the project	c) Communication between team	d) Maintaining the project
2.	The second step in creating a project is _____.			
	a) Defining the project	b) Managing the project	c) Communication between team	d) Maintaining the project
3.	The third step in creating a project is _____.			
	a) Defining the project	b) Managing the project	c) Communication between team	d) Maintaining the project
4.	A ____ is a small unit of work or a <u>specific activity</u> that needs to be completed as part of a project or process.			
	a) Task	b) Duration	c) Critical path	d) Project
5.	A _____ is an activity that has a start and a finish .			
	a) Task	b) Duration	c) Critical path	d) Project
6.	_____ is a chart that illustrates the <u>project's tasks</u> and their durations against a calendar .			
	a) Gantt chart	b) PERT/CPM	c) WBS	d) NPV
7.	A _____ is a hierarchical decomposition of the work to be executed by the project team to accomplish projects objectives and create the required deliverables.			
	a) WBS	b) PERT	c) Effort	d) None
8.	A _____ is an essential activity that directly affects (impacts) <i>the success or timely completion</i> of a project.			
	a) Gantt chart	b) WBS	c) Critical path	d) Critical Task
9.	The _____ is the longest sequence of dependent tasks in a project that determines the <u>shortest possible duration to complete the project</u> .			
	a) Gantt chart	b) WBS	c) Critical path	d) Critical Task
10.	_____ is the path having all activities with zero slack .			
	a) Gantt chart	b) WBS	c) Critical path	d) Critical Task
11.	A chart representing <u>project tasks</u> against a calendar in a horizontal bar format is known as _____.			
	a) Milestone	b) Goal	c) Gantt chart	d) PERT chart
12.	_____ is a horizontal bar chart that shows <u>project tasks</u> against a calendar .			
	a) Milestone	b) Goal	c) Gantt chart	d) PERT chart
13.	A _____ is used to show a <u>project's tasks</u> , durations, and dependencies visually.			
	a) Milestone	b) Task	c) Gantt chart	d) Custom Field



14.	WBS stands for _____.							
	a)	Work Breakdown Schedule						
	b)	Work Breakdown Structure						
	c)	Wide Business Strategy						
	d)	Work-based Solution						
15.	FF dependency implies _____							
	a)	Task 2 must finish before Task 1 can start						
	b)	Task 1 must finish before Task 2 can start						
	c)	The work can overlap or proceed in parallel.						
	d)	The work can happen in parallel, and the tasks must end at the same time.						
16.	FS dependency implies _____							
	a)	Task 1 must finish before Task 2 can start						
	b)	The work can overlap or proceed in parallel.						
	c)	The work can happen in parallel, and the tasks must end at the same time.						
	d)	This is not possible						
17.	SS dependency implies _____							
	a)	Task 2 must finish before Task 1 can start						
	b)	Task 1 must finish before Task 2 can start						
	c)	The work can overlap or proceed in parallel, and the tasks must start at the same time.						
	d)	The work can happen in parallel, and the tasks must end at the same time.						
18.	SF dependency implies _____							
	a)	Task 1 must finish before Task 2 can start						
	b)	Task 1 must start before Task 2 can finish						
	c)	The work can overlap or proceed in parallel.						
	d)	The work can happen in parallel, and the tasks must end at the same time.						
19.	A _____ refers to a task that is interrupted and resumed later in a project.							
	a)	Summary	b)	Split Task	c)	Milestone	d)	WBS
20.	When a person has a to halt work for a specific time and <u>continue</u> it again, we use _____ in Gantt chart.							
	a)	Summary	b)	Split Task	c)	Milestone	d)	WBS
21.	A _____ is a significant event or achievement in a project that marks the completion of a key phase or task.							
	a)	Summary Task	b)	Milestone	c)	Split Task	d)	Recurring Task
22.	If we need an indication that a task is finished , we add a _____ in MS Project.							
	a)	Summary Task	b)	Milestone Task	c)	Split Task	d)	Recurring Task
23.	A _____ is a task that repeats regularly over a set period, such as daily, weekly, or monthly.							
	a)	Summary Task	b)	Milestone Task	c)	Split Task	d)	Recurring Task



24.	If we need to add a <i>weekly meeting</i> throughout the project's timeline we use _____.			
	a) Summary Task	b) Milestone Task	c) Split Task	d) Recurring Task
25.	A _____ is a high-level task holding subtasks to help organizing the tasks list.			
	a) Summary Task	b) Subtasks	c) Milestone task	d) None
26.	_____ is a task holding subtasks to help organizing the tasks list.			
	a) D6 Summary task	b) Subtasks	c) Milestone tasks	d) None
27.	What are the types of task links in MS project?			
	a) Finish to start link	b) Start to start link	c) Finish to finish link	d) All of the above
28.	Deliverables (milestones) are represented in Gantt chart as _____.			
	a) Bars	b) Circles	c) Triangles	d) Diamonds
29.	Tasks are represented in Gantt chart as _____.			
	a) Bars	b) Circles	c) Triangles	d) Diamonds
30.	What is the default timescale setting of the Gantt chart view in MS project?			
	a) One tier	b) Two tiers	c) Three tiers	d) Four tiers
31.	In MS Project, how can you recognize project milestone ?			
	a) By applying boldface	b) By indenting that task	c) By assigning a duration of zero for that task	d) None
32.	_____ refers to the delay (waiting time) between the completion of one task and the start of the next task in a project.			
	a) Lag	b) Lead	c) Slack	d) None
33.	A positive time interval between activities is known as _____ time.			
	a) Lag	b) Lead	c) Slack	d) None
34.	A negative time interval between activities is known as _____ time.			
	a) Lag	b) Lead	c) Slack	d) None
35.	_____ is the amount of time an <i>activity can be delayed without affecting the end date</i> of a project.			
	a) Lag	b) Lead	c) Slack	d) None
36.	_____ is computed as the difference between the latest and earliest start of an activity or, equivalently, between the latest and earliest end .			
	a) Lag	b) Lead	c) Slack	d) None
37.	In MS project, we manage resources needed for work in _____.			
	a) CPM	b) PERT	c) Gantt Chart	d) Resource Sheet
38.	Resource types in MS project are _____.			
	a) Work	b) Material	c) Cost	d) All of them
39.	In MS project, we assign resources to tasks using _____.			
	a) Manually schedule	b) Inactivate Tasks	c) Inspect	d) Assign Resources



40.	In MS Project, Resources can be shown in _____ column.			
	a) Start date	b) Finish date	c) Predecessors	d) Resource Names
41.	As soon as possible (ASAP) and As late as possible (ALAP) are _____ constraints			
	a) Flexible	b) Semi-flexible	c) Inflexible	d) None
42.	Start no earlier than (SNET), Finish no earlier than (FNET), start no later than (SNLT), finish no later than (FNLT) are _____ constraints.			
	a) Flexible	b) Semi-flexible	c) Inflexible	d) None
43.	Must start on (MSO), Must finish on (MFO) are examples on _____ constraints.			
	a) Flexible	b) Semi-flexible	c) Inflexible	d) None
44.	Which of the following is a type of cost?			
	a) Fixed cost	b) Variable cost	c) Expensive cost	d) A & B
45.	We use _____ to print reports on project summary, costs, assignments, critical tasks or etc.			
	a) Reports	b) Resource Sheet	c) Gantt Chart	d) WBS
46.	A _____ is a user-defined data field used to track specific information <i>not included in standard fields</i> .			
	a) Custom Field	b) Predecessor	c) Gantt Chart	d) Task
47.	A _____ represents the visual schedule of tasks and events with their start and end dates.			
	a) Timeline	b) Working Area	c) Milestone	d) Recurring Task
48.	The _____ in project management refers to the physical or virtual space where tasks are carried out.			
	a) Working Area	b) Custom Field	c) Gantt Chart	d) Summary Task
49.	_____ are tasks that <i>must be completed before</i> the next task can begin in a project.			
	a) Summary Task	b) Predecessors	c) Split Tasks	d) Milestones
50.	A _____ is a limitation or restriction that affects project planning or execution.			
	a) Task	b) Constraint	c) Critical Path	d) Recurring Task
51.	We can add a any field to the Gantt chart in MS project using _____.			
	a) Insert Task	b) Insert Column	c) Insert row	d) Inactivate Task
52.	We can add a custom field to the Gantt chart in MS project using _____.			
	a) Custom Fields	b) Milestone	c) Recurring Tasks	d) Resource Sheet



الاحتياط واجب (مسائل نظري)

Q: What are the deployment types?

A: Cutover, Parallel, Piloting, Phased

Q: What is the triple constraint triangle?

A: Quality, Time, Cost

Q: What are the testing types?

A: Unit testing, Integration testing, System testing, Usability testing

Q: What are the integration types?

A: Horizontal and vertical

Q: What is the main purpose of WBS?

A: Determine the scope of project

Q: What is the difference between WBS and Gantt Chart?

A: WBS Shows the hierarchal relationship and Gantt Chart shows the timeline.

Q: What are the WBS rules?

A: 100% rule, Mutually exclusive rule, No need to be balanced rule.

Q: What is the approach of WBS that start from an existing WBS?

A: Analogy Approach

Q: What are the WBS types?

A: Process WBS, Product WBS, Hybrid WBS, etc.

Q: What is the main purpose of estimation?

A: Budget, Timeline, Resources

Q: What is the effort?

A: **Amount of work** required to complete an activity.

Q: What is a consequence of optimistic?

A: Project suffer from delaying.

